

Product Feedback Analyzer using MongoDB (Console-Based REST API Project)

Objective

The Product Feedback Analyzer is a console-based backend application designed to collect, manage, and analyse product feedback from users. The system allows users to submit feature requests, bug reports, and improvement suggestions while enabling product managers to review, prioritize, and track feedback. The project demonstrates CRUD operations, REST API development, MongoDB document relationships, indexing, text search, and aggregation pipelines.

Technologies Used

- Runtime Environment: Node.js
- Backend Framework: Express.js
- Database: MongoDB
- API Testing Tools: Postman / Swagger
- Version Control: Git & GitHub
- Development Type: Console-Based Backend Application

System Entities

1. Users
2. Feedback
3. Votes
4. Categories
5. Status Tracking

Working of the System

1. Users create an account and log in.
2. Users submit feedback in the form of feature requests, bug reports, or improvement suggestions.
3. The feedback is stored in MongoDB with details such as title, description, category, creator, submission date, and status.
4. Users can browse all existing feedback through API endpoints.
5. Search and filter operations allow feedback retrieval based on category, status, or keywords.

6. Users can upvote existing feedback instead of submitting duplicate requests.
7. Vote records are stored and linked to both the user and the feedback item.
8. Product managers access administrative APIs to review submitted feedback.
9. Product managers update feedback status as Open, Under Review, Planned, In Progress, or Completed.
10. MongoDB Aggregation Pipelines process stored data to generate analytics such as:
 - Most requested features
 - Most common feedback categories
 - Open vs Completed requests
 - Monthly feedback trends
11. Analytical insights help product teams prioritize product development based on actual user demand.
12. Users can track the progress and current status of their submitted feedback.

MongoDB Operations Implemented

- Create User
- Submit Feedback
- View Feedback
- Update Feedback
- Delete Feedback
- Search Feedback
- Upvote Feedback
- Filter Feedback
- Generate Analytics
- Aggregation Queries
- Text Search Indexing
- Complete CRUD Operations

MongoDB Relationships

- One User → Many Feedback Submissions
- One Feedback → Many Votes

- One Category → Many Feedback Items
- Many Users → Many Feedback Items (through voting)

REST API Workflow

User Request → Express.js REST API → Business Logic Layer → MongoDB Database → Aggregation & Analysis → API Response → Console/Postman Output

Expected Outcome

The system provides a centralized platform for collecting, organizing, and analyzing customer feedback. By utilizing MongoDB's flexible document structure, indexing features, and aggregation framework, product teams can identify customer priorities, monitor feedback trends, and make informed product decisions efficiently.